

Can headlines alone provide enough information for accurate and efficient news classification?

Team 096: Shufa Wu

Questions

- (1) Can a machine learning model accurately classify news articles using only the headline rather than the full article text?
- (2) How much predictive information is contained in a headline, and which NLP features contribute most to classification performance?

Introduction & Problem Statement

With the rapid growth of online news platforms, thousands of news articles are published daily across a wide range of topics, including politics, business, sports, entertainment, science, and technology. To organize this large volume of information, news organizations rely on automatic news categorization systems that enable efficient content management, personalized recommendations, and improved search capabilities. Most traditional news classification systems rely on the full text of news articles, as the complete document provides rich semantic and contextual information for classification (Kowsari et al., 2019). However, processing full-length articles requires additional preprocessing and larger feature representations, resulting in greater computational and memory requirements than processing shorter text such as headlines. In many practical applications, such as real-time news aggregation, personalized recommendation systems, and mobile news platforms, these computational costs may limit scalability and increase prediction latency. These challenges motivate investigating whether comparable classification performance can be achieved using only article headlines, which are substantially shorter and less computationally expensive to process.

This project focuses on whether news headlines alone contain enough semantic information to accurately classify articles into their respective categories. By restricting the input to headline text, the proposed approach substantially reduces the amount of text that must be processed, thereby decreasing preprocessing complexity and computational requirements. The study will compare three supervised machine learning algorithms to determine whether competitive classification performance can be achieved using this lightweight representation.

The project will utilize the HuffPost News Category Dataset from Kaggle (Misra, 2022), which contains more than

200,000 news articles across 42 categories. Although the dataset includes additional information such as short descriptions, publication dates, authors, and URLs, only the headline text will be used as the input feature. This experimental design isolates the predictive value of headlines and enables a direct evaluation of the trade-off between computational efficiency and classification performance.

Methodology

Data Exploration: The project will begin with data preparation and exploratory data analysis (EDA). The HuffPost News Category Dataset will be examined for missing values, duplicate records, and class imbalance across the 42 news categories. Exploratory analyses will include the distribution of headline lengths, vocabulary richness, and the frequency of shared keywords among semantically similar categories (e.g., POLITICS vs. WORLD NEWS and WELLNESS vs. HEALTHY LIVING). These analyses will guide preprocessing decisions and provide insight into the classification difficulty of individual categories using headline text alone.

Natural Language Processing: The preprocessing pipeline will consist of lowercase conversion, punctuation removal, tokenization, stop-word removal, and lemmatization (HaCohen-Kerner et al., 2020). Lowercasing ensures that words differing only by capitalization are treated as identical tokens, while punctuation removal reduces unnecessary noise. Tokenization separates headlines into individual words, stop-word removal eliminates common words with little predictive value, and lemmatization converts words to their base forms, reducing vocabulary size while preserving semantic meaning.

Following preprocessing, headlines will be transformed into numerical feature vectors using the Term Frequency–Inverse Document Frequency (TF-IDF) representation (Kowsari et al., 2019). TF-IDF assigns greater weight to informative terms while reducing the influence of frequently occurring words across the corpus. Because headlines are concise and information-dense, TF-IDF is well suited for emphasizing discriminative keywords while producing feature representations that are computationally efficient to process.

Model Development: Three supervised machine learning algorithms will be implemented and compared to evaluate the effectiveness of headline-based news classification. **Multi-nomial Naïve Bayes** will serve as a computationally effi-

cient baseline due to its simplicity and strong performance on sparse text classification tasks. **Logistic Regression** will provide a robust linear classifier while offering interpretable feature coefficients that facilitate analysis of the linguistic characteristics associated with each news category. **Support Vector Machine (SVM)** will serve as the primary classification model because of its proven effectiveness on high-dimensional, sparse feature spaces such as TF-IDF representations and its consistently strong performance in document classification.

The dataset will be divided into training and testing sets using an 80/20 stratified split to preserve the class distribution across all 42 categories. Hyperparameters for each model will be optimized using grid search in conjunction with five-fold cross-validation to identify the best-performing configuration while reducing the risk of overfitting. Model performance will be evaluated using accuracy, precision, recall, and macro-averaged F1-score. In addition to predictive performance, preprocessing time, model training time, prediction time, and TF-IDF vocabulary size will be recorded to quantify the computational efficiency of the proposed headline-based classification approach.

Model Interpretability: While predictive performance is the primary objective, understanding how a model achieves its predictions provides valuable insight into the effectiveness of headline-based classification. Therefore, this project will incorporate an interpretability analysis to examine the linguistic features that contribute most strongly to each news category.

Following model training, the highest-weighted TF-IDF terms associated with several representative news categories were examined to identify the words that contributed most strongly to distinguishing each category from the others. For example, terms such as *NBA*, *playoffs*, and *season* are expected to strongly indicate the *Sports* category, whereas words including *Congress*, *Senate*, and *election* may be highly associated with *Politics*. Similarly, terms such as *startup*, *artificial intelligence*, and *software* are expected to characterize articles in the *Technology* category. Examining these feature weights provides transparency by revealing how the classifier distinguishes among categories and offers qualitative evidence that the model learns meaningful semantic relationships rather than relying on spurious correlations.

Furthermore, the interpretability analysis will be used to investigate common sources of classification error. Categories with highly overlapping vocabularies may exhibit greater classification confusion, providing insight into the inherent limitations of headline-only classification. By relating influential keywords to model performance, this analysis will help determine whether news headlines alone contain sufficient discriminative information to support accurate and computationally efficient news categorization.

Evaluation Strategy

Model performance will be evaluated using Accuracy, Precision, Recall, and Macro F1-score. Because the dataset contains 42 news categories with varying sample sizes, Macro F1-score will serve as the primary evaluation metric, as it assigns equal importance to each category and provides a more balanced assessment of multi-class classification performance.

In addition to predictive performance, the computational efficiency of the proposed headline-based approach will be evaluated by recording preprocessing time, model training time, prediction time, and TF-IDF vocabulary size. These measurements will be used to quantify the computational efficiency of headline-based classification and assess whether competitive predictive performance can be achieved using substantially less textual information.

Confusion matrices will be analyzed to identify categories that are most frequently misclassified. Rather than solely reporting aggregate performance metrics, this analysis will investigate why certain categories are more difficult to distinguish using headlines alone. For example, categories such as *Politics* and *World News* are expected to exhibit greater confusion because of their overlapping vocabulary, whereas categories such as *Sports* and *Technology* may be more easily distinguished due to their domain-specific terminology.

Finally, the highest-weighted TF-IDF features learned by the Logistic Regression classifier will be examined to interpret the model's predictions. Identifying the most influential terms for each news category will provide qualitative evidence that the classifier captures meaningful semantic patterns and will complement the quantitative evaluation by explaining the linguistic characteristics that drive classification decisions.

Results

This section presents the performance of the three machine learning models evaluated for news headline classification: Multinomial Naive Bayes, Logistic Regression, and Linear Support Vector Machine (Linear SVM). All models were trained using TF-IDF representations of the preprocessed headlines and optimized through five-fold cross-validation using GridSearchCV. Model performance was evaluated using classification accuracy, macro-averaged precision, recall, and F1-score. In addition, computational efficiency was assessed by measuring model training and prediction times.

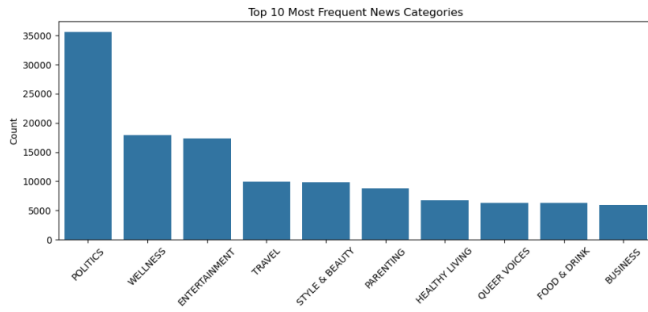


Figure 1: Distribution of News Categories (Top 10)

Among the three models, Linear SVM achieved the highest overall classification accuracy (58.2%) while also obtaining the highest macro F1-score (0.45). Logistic Regression achieved nearly identical macro F1 performance (0.44) but achieved slightly lower classification accuracy (57.4%). Multinomial Naive Bayes achieved the lowest overall performance, with an accuracy of 50% and a macro F1-score of only 0.26.

Although Naive Bayes demonstrated the highest macro precision (0.62), its macro recall was substantially lower (0.23), indicating that the model generated conservative predictions for many categories. As a result, numerous minority classes were rarely predicted, resulting in poor recall and consequently a low macro F1-score. In contrast, both Logistic Regression and Linear SVM achieved a much better balance between precision and recall, leading to considerably higher macro F1 values.

The overall results demonstrate that linear discriminative classifiers are more effective than the probabilistic Naive Bayes model for multiclass news headline classification using TF-IDF features. The improvement is likely due to the ability of Logistic Regression and Linear SVM to learn more flexible decision boundaries within the high-dimensional sparse feature space generated by TF-IDF.

	Model	Best Hyperparameter	Accuracy	Macro Precision	Macro Recall	Macro F1	Training Time (s)	Prediction Time (s)
0	Naive Bayes	$\alpha = 0.1$	0.500	0.62	0.23	0.26	2.25	0.040
1	Logistic Regression	$C = 10$	0.580	0.56	0.40	0.44	207.61	0.040
2	Linear SVM	$C = 1$	0.582	0.49	0.43	0.45	55.03	0.046

Figure 2: Model Performance Comparison

Hyperparameter Optimization: Hyperparameter tuning was performed using five-fold cross-validation. For Multinomial Naive Bayes, the smoothing parameter was optimized over several candidate values, with $\alpha = 0.1$ producing the highest validation performance. Logistic Regression selected a regularization parameter of $C = 10$, while Linear SVM achieved its best performance using the default regularization parameter of $C = 1$.

The cross-validation macro F1-scores were 0.381 for Naive Bayes, 0.441 for Logistic Regression, and 0.440 for Linear SVM. These values closely matched the corresponding test-set macro F1-scores, suggesting that the models generalized well to unseen data and exhibited little evidence of overfitting.

Hyperparameter tuning noticeably improved the performance of Naive Bayes, whereas only marginal improvements were observed for Logistic Regression and Linear SVM. This indicates that the two linear classifiers are relatively robust to moderate changes in the regularization parameter, while Naive Bayes is more sensitive to the smoothing parameter.

Computational Efficiency: In addition to predictive performance, computational efficiency was evaluated by measuring both model training time and prediction time.

Naive Bayes required only 2.25 seconds to complete model training, making it the fastest classifier by a substantial margin. Linear SVM required approximately 55 seconds for training, whereas Logistic Regression required over 207 seconds, making it the most computationally expensive model in this study. Despite these differences in training cost, prediction times were nearly identical across all three models, requiring approximately 0.04 seconds to classify the entire test set.

The computational results indicate an important trade-off between efficiency and predictive performance. While Naive Bayes trained almost twenty-five times faster than Linear SVM and nearly one hundred times faster than Logistic Regression, this speed came at the expense of substantially lower classification performance. Linear SVM achieved the highest predictive accuracy while requiring only about one-quarter of the training time of Logistic Regression, providing the best balance between computational cost and predictive capability.

Category-Level Classification Performance: Performance varied considerably across the forty-two news categories. Categories with highly distinctive vocabulary generally achieved the highest F1-scores. For example, Politics, Style Beauty, Weddings, Travel, and Entertainment consistently achieved F1-scores above 0.68 when classified using Linear SVM. These categories contain highly specific terminology that is well captured by the TF-IDF representation.

In contrast, categories such as Arts, Taste, U.S. News, Healthy Living, and WorldPost produced considerably lower F1-scores. These categories frequently share vocabulary with semantically similar topics, making them more difficult to distinguish based solely on headline text. For example, headlines discussing wellness may overlap substantially with those concerning healthy living, while political news frequently shares terminology with U.S. News and World News. The models rely only on headline text and do not utilize article content or contextual information, these semantic overlaps increase the likelihood of misclassification.

The relatively low macro F1-score compared with overall accuracy also reflects the class imbalance present in the dataset.

Large categories such as Politics and Entertainment contain substantially more training examples than many smaller categories, allowing the classifiers to learn more representative decision boundaries for those classes. Minority categories, on the other hand, contributed disproportionately to the reduction in macro-averaged performance metrics.

Feature Analysis:

Table 1: Overall Performance of the Classification Models	
Category	Top 10 Keywords
Politics	trump, democrat, gop, obama, republican, senate, clinton, biden, congress, senator
Sports	nfl, nba, sport, football, mlb, lebron, fan, soccer, olympic, player
Business	business, uber, ceo, wall street, marketing, worker, job, company, krugman, woman business
Technology	apple, google, facebook, iphone, microsoft, twitter, hacker, Samsung, amazon, tech
Entertainment	trailer, box office, movie, kardashian, netflix, snl, oscar, adele, clip, jenner

To better understand the representation of learned features, the highest TF-IDF weighted keywords were examined for several representative news categories. Table 1 presents the ten most informative terms for Politics, Sports, Business, Technology, and Entertainment.

The extracted keywords closely align with their corresponding news topics. For example, the Politics category is characterized by terms such as Trump, Democrat, Republican, Senate, and Congress, while the Technology category is dominated by Apple, Google, Facebook, iPhone, and Microsoft. Similarly, Sports is associated with terms including NFL, NBA, football, and Olympic, whereas Entertainment is characterized by words such as movie, Netflix, Oscar, and trailer.

The semantic relevance of these keywords demonstrates that the TF-IDF representation successfully captures discriminative vocabulary for each news category. Rather than relying on common words shared across documents, the models primarily learned category-specific terms that effectively distinguish one news topic from another. This observation also explains the strong performance of the linear classifiers, which exploit these discriminative features to construct effective decision boundaries within the high-dimensional TF-IDF feature space.

Confusion Matrix: Figure 3 demonstrates the normalized confusion matrix for the ten most frequent news categories using the Linear SVM classifier. Most categories exhibit strong diagonal values, indicating that the majority of headlines are correctly classified. Categories with distinctive vocabularies, including Politics (93%), Style Beauty (87%), Travel (84%), and Entertainment (84%), achieve particu-

larly high classification accuracy. In contrast, Healthy Living demonstrates substantial correlation with Wellness, with approximately 61% of Healthy Living headlines being predicted as Wellness. This result suggests that these two categories share considerable lexical overlap, making them difficult to distinguish using headline text alone. Similar, though less pronounced, confusion is observed between Business and Politics, reflecting overlap in economic and policy-related terminology. Overall, the confusion matrix indicates that classification errors primarily occur between semantically related categories rather than unrelated topics.

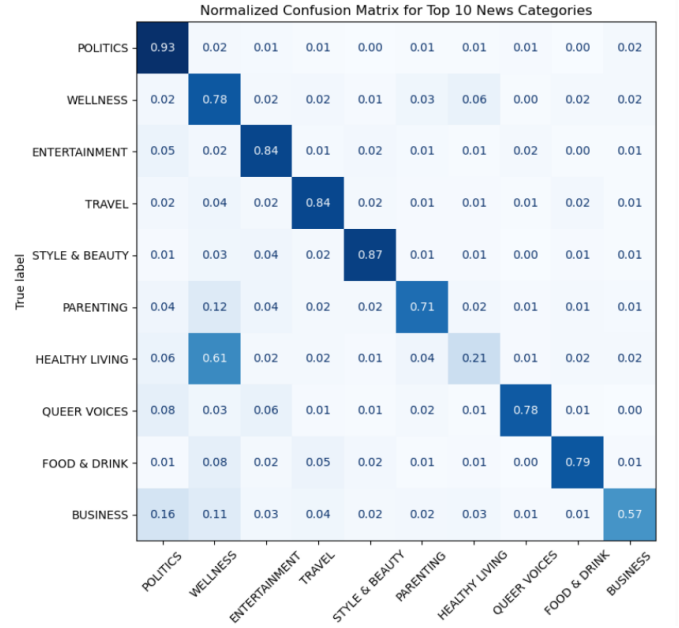


Figure 3: Confusion Matrix for Top 10 News Categories

Conclusion: The results demonstrate that traditional machine learning algorithms combined with TF-IDF feature representations provide an effective solution for multiclass news headline classification. Among the three models evaluated, Linear Support Vector Machine (Linear SVM) achieved the best overall performance, obtaining the highest test accuracy (58.2%) and macro F1-score (0.45) while maintaining substantially lower training time than Logistic Regression. Although Multinomial Naive Bayes required the least computational effort, its predictive performance was considerably lower, indicating that faster training alone does not necessarily translate into better classification quality.

The analysis also highlights the importance of evaluating models using multiple performance metrics. While overall accuracy provides a general measure of predictive performance, the macro-averaged metrics reveal the impact of class imbalance across the 42 news categories. Linear SVM and Logistic Regression produced a more balanced trade-off between precision and recall than Naive Bayes, resulting in higher macro F1-scores and better performance on minority

classes.

Feature analysis further demonstrated that the TF-IDF representation successfully captured meaningful category-specific vocabulary. Highly weighted terms such as Trump, NFL, Apple, and Netflix aligned closely with their corresponding news categories, providing qualitative evidence that the learned features were both interpretable and semantically meaningful. The confusion matrix further showed that most classification errors occurred between semantically related categories, particularly Healthy Living and Wellness, suggesting that the remaining errors were primarily caused by overlapping vocabulary rather than random model behavior.

Overall, these findings indicate that Linear SVM offers the best balance between predictive performance, computational efficiency, and model interpretability for headline-based news classification. The results also demonstrate that accurate topic classification can be achieved using only news headlines, although additional contextual information may be required to improve performance for categories with highly similar language.

Limitations and Future Work

Several limitations should be acknowledged. First, the dataset consists exclusively of English-language news headlines collected from the HuffPost News Category Dataset. Consequently, the findings may not generalize to news published by other media outlets, different writing styles, or multilingual news sources. Second, headlines are intentionally brief summaries of news articles and therefore often lack the contextual information contained in the full text. This limitation makes it more difficult to distinguish between categories with similar vocabulary, such as Politics, U.S. News, and World News, which can reduce classification performance.

Another limitation is the class imbalance present in the dataset. Categories such as Politics and Entertainment contain substantially more training examples than many smaller categories. As a result, the models generally achieved better performance on majority classes while producing lower recall and F1-scores for minority categories.

Finally, this project evaluated traditional machine learning algorithms using TF-IDF feature representations rather than modern deep learning approaches. Although TF-IDF combined with linear classifiers provides an interpretable and computationally efficient solution, these methods cannot capture contextual relationships between words as effectively as recent neural language models.

Future work could include several extensions to improve classification performance. First, incorporating additional textual information, such as article summaries or release dates, may provide richer semantic context and improve the classification of categories with overlapping vocabulary. Second, transformer-based language models such as BERT could be explored to determine whether contextual embeddings

lead to higher predictive performance than TF-IDF features. Finally, future work could examine techniques for addressing class imbalance, including class weighting, oversampling, or data augmentation, to improve performance on underrepresented news categories without sacrificing overall classification accuracy.

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